PROJECT PROPOSAL



PM2.5 Air Quality Prediction for the Mae Fah Luang University

PHYO THANT KYAW

PHONE WAI YAN THAING

KYAW PHONE THU

KYAW THURA

BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2026

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

PM2.5 Air Quality Prediction for the Mae Fah Luang University

PHYO THANT KYAW
PHONE WAI YAN THAING
KYAW PHONE THU
KYAW THURA

A **SENIOR PROJECT** SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN **COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY
2026
© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

PM2.5 Air Quality Prediction for the Mae Fah Luang University

PHYO THANT KYAW
PHONE WAI YAN THAING
KYAW PHONE THU
KYAW THURA

THIS **SENIOR PROJECT** HAS BEE APPROVED
TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING
2026

EXAMINING COMMITTEE

Dr. Khwunta Kirimasthong - ADVISOR

Asst. Prof. Gp. Capt. Dr. Thongchai Yooyativong - COMMITTEE

Dr. Paweena Suebsombut - COMMITTEE

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advices and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

PHYO THANT KYAW

PHONE WAI YAN THAING

KYAW PHONE THU

KYAW THURA

Title: **PM2.5 Air Quality Prediction for the Mae Fah Luang University**

Author PHYO THANT KYAW
PHONE WAI YAN THAING
KYAW PHONE THU
KYAW THURA

Degree Bachelor of Engineering
(Computer Engineering)

Supervisory Committee

- **Dr. Khwunta Kirimasthong** - Advisor
- **Asst. Prof. Gp. Capt. Dr.Thongchai Yooyativong** - Committee
- **Dr. Paweena Suebsombut** - Committee

Abstract

Northern Thailand faces a severe annual haze crisis, worsened by "Temperature Inversion" in valley basins like Mae Fah Luang University (MFU). Existing global air quality models rely on coarse grids that fail to capture these hyper-local micro-climates, resulting in delayed and generalized warnings that leave the campus community vulnerable to sudden pollution spikes. To address this gap, this project develops a localized PM2.5 forecasting framework specifically calibrated for the MFU topography.

The system evaluates a comprehensive 4-way machine learning framework—featuring Extreme Gradient Boosting **(XGBoost**) as the champion model (achieving an peak accuracy **R^2 of 86.52%),** alongside Random Forest, Support Vector Regression (SVR), and Multiple Linear Regression (MLR). These models are trained on a reliable 7-year historical dataset from Chiang Mai, leveraging identical regional valley physics. For real-time predictions, the architecture integrates the OpenWeatherMap API to dynamically fetch live and 5-day forecasted meteorological data using MFU's exact GPS coordinates.

The final product is a proactive web dashboard deployed via the Streamlit framework. This interactive platform features real-time AQI updates, historical trend analysis, and a 4way "Live Model Showdown" comparing both advanced ensemble engines and linear baseline models on the fly. Ultimately, this tool translates complex machine learning outputs into clear, actionable health guidelines, empowering the MFU community with hyper-local, 5-day predictions.

Keywords: PM2.5 Prediction, XGBoost, Streamlit Dashboard, Ensemble Machine Learning, OpenWeatherMap API, Valley Basin.

CONTENTS

ACKNOWLEDGEMENT iii

Abstract iv

LIST OF TABLES **vii**

LIST OF FIGURES **viii**

CHAPTER 1 INTRODUCTION **1**

1.1 Background and Rationale 1

1.2 Objective 1

1.3 Scope 2
1.4 Methodology 3
1.5 Plan 4
1.6 Expected Result **Error! Bookmark not defined.**
1.7 Place and Equipment 5
CHAPTER 2 LITERATURE REVIEW **8**
2.1 Related Theory 8
2.2 Related Research 8
2.3 Related Work 9
2.4 Related Technology 9
CHAPTER 3 ANALYSIS AND DESIGN **11**
3.1 Existing System 11
3.2 New System 11
3.3 Use case diagram 12
3.4 Activity diagram 13
CHAPTER 4 CONCLUSION **15**
4.1 Conclusion 15
4.2 Discussion 15
REFERENCES **16**

LIST OF TABLES

Table **Page**

Table 1.1 Working plan 4

LIST OF FIGURES

Figure **Page**

Figure 3.1: Use case diagram of the existing system 8

Figure 3.2**:** System Architecture of New System 9
Figure 3.3: Use case diagram of the new system 9
Figure 3.4: Activity diagram of the new system 10
Figure 3.5: Entity Relationship Diagram(ERD) 10

CHAPTER-1 INTRODUCTION

1.1 Background and Rationale

Northern Thailand, particularly the provinces of Chiang Rai and Chiang Mai, endures a severe annual air pollution crisis characterized by hazardous levels of PM2.5. According to the Pollution Control Department (PCD) in their "Historical Air Quality Data 2016-2023" report, this phenomenon is significantly exacerbated by the region's unique "Valley Basin" geography. As explained by C.D. Whiteman in Mountain Meteorology: Fundamentals and Applications, these bowl-shaped terrains facilitate a meteorological condition known as a "Temperature Inversion" during the dry season. This inversion layer acts as a physical "lid," trapping smoke and particulate matter near the ground and preventing the vertical dispersion of pollutants.

Current reliance on global or regional air quality models presents significant challenges for local populations. As noted by IQAir regarding their AirVisual platform, while these global tools are excellent for general awareness, they often rely on coarse grid resolutions approximately 9km x 9km which are too broad to capture the specific microclimates of small valleys like the one housing Mae Fah Luang University (MFU). Furthermore, satellite-based estimations often suffer from blind spots during nighttime or heavy cloud cover, leading to delayed or generalized warnings.

In response to these localized environmental challenges, this work develops a specialized data-driven framework to bridge the gap between global monitoring and hyperlocal forecasting. Because local, continuous 7-year datasets are highly fragmented, this project strategically utilizes validated meteorological data from Chiang Mai—a region sharing the exact same Northern Thailand micro-climate physics—to train an advanced Machine Learning engine. The final output is a proactive, on-the-fly PM2.5 forecasting system specifically calibrated for the MFU valley, deployed as an interactive web application built on the Streamlit framework. Unlike reactive tools, this system integrates live API data to provide a 5-day predictive forecast, enabling the MFU community to make informed, healthconscious decisions before hazardous exposure occurs.

1.2 Objective

- **To establish a real-time data infrastructure:** To integrate the OpenWeatherMap API for fetching live and 5-day forecasted meteorological data specific to MFU's GPS coordinates, replacing the reliance on static local databases for real-time inference.

- **To implement and evaluate advanced predictive modeling**: To develop, train, and rigorously compare four distinct Machine Learning models—**Extreme Gradient Boosting (XGBoost), Random Forest, Support Vector Regression (SVR), and Multiple Linear Regression (MLR)**—to accurately forecast PM2.5 levels based on 8 key weather and historical parameters.

- **To deploy a user-centric predictive dashboard:** To build a dynamic, 3-tab web application using the Streamlit framework that visualizes real-time status, historical trends, and live model showdown comparisons for the university community.

1.3 Scope of work

This project is defined by the following technical boundaries, data parameters, and functional requirements:

1.3.1 Application Type

- **Web-Based Interface:** The final product is a Web Application developed using the Streamlit framework.

- **Local Deployment & Access:** For this current phase of the project, the system is deployed and hosted on a local server environment (localhost). It is fully accessible via any modern web browser and utilizes responsive web design to ensure proper visualization and functionality across different screen sizes.

1.3.2 Data Sources

This project fundamentally shifts from static local databases to a dynamic, API-driven architecture, utilizing two primary data streams:

- **Historical Training Data (Kaggle):** To ensure model reliability, 7 years of continuous, pre-cleaned meteorological and PM2.5 data from Chiang Mai is utilized. This source is chosen over highly fragmented local datasets because it provides validated meteorological infrastructure while sharing the identical Northern Thailand micro-climate physics.

- **Live Inference Data (OpenWeatherMap API):** For real-time predictions, the system fetches live and 5-day forecasted weather data using the exact GPS coordinates of MFU. This acts as the fresh input for the pre-trained machine learning engine.

1.3.3 Data Modeling and Dataset

- **Dataset Timeframe:** 7 years of historical data to ensure the model captures recurring seasonal "burning season" trends.

- **Modeling Technique:** Supervised Machine Learning utilizing a comprehensive 4way comparative framework: **XGBoost** as the primary champion engine, **Random Forest** as the secondary ensemble, alongside Support Vector Regression (**SVR**) and Multiple Linear Regression (**MLR**) serving as structural baselines to scientifically demonstrate the complex, non-linear nature of valley weather.

- **Target Variable: PM2.5 Concentration**.

- **Independent Variables (Features):** 8 key features comprising 7 meteorological factors (Temperature, Humidity, Wind Speed, Wind Direction, Atmospheric Pressure, Precipitation, Sunshine) and 1 historical anchor factor (pm25_lag1).

1.3.4 System Functional Requirements by User Role

- **General Users:** Can view real-time local PM2.5 levels, access a 5-day predictive trend chart, and receive dynamic health guidelines based on Air Quality Indices.

- **System Administrator:** Can monitor API connection statuses, view the Live Model Showdown in the backend analysis tab, and manage the Streamlit deployment environment.

1.4 Methodology

- **Problem Identification & Literature Review:** Analyzed the limitations of Global Grid Models and the specific meteorological challenges of the MFU Valley (Temperature Inversion).

- **Data Engineering (Phase 1):** Sourced and pre-processed 7 years of validated historical data from Chiang Mai, structuring the 8 key meteorological features for machine learning input.

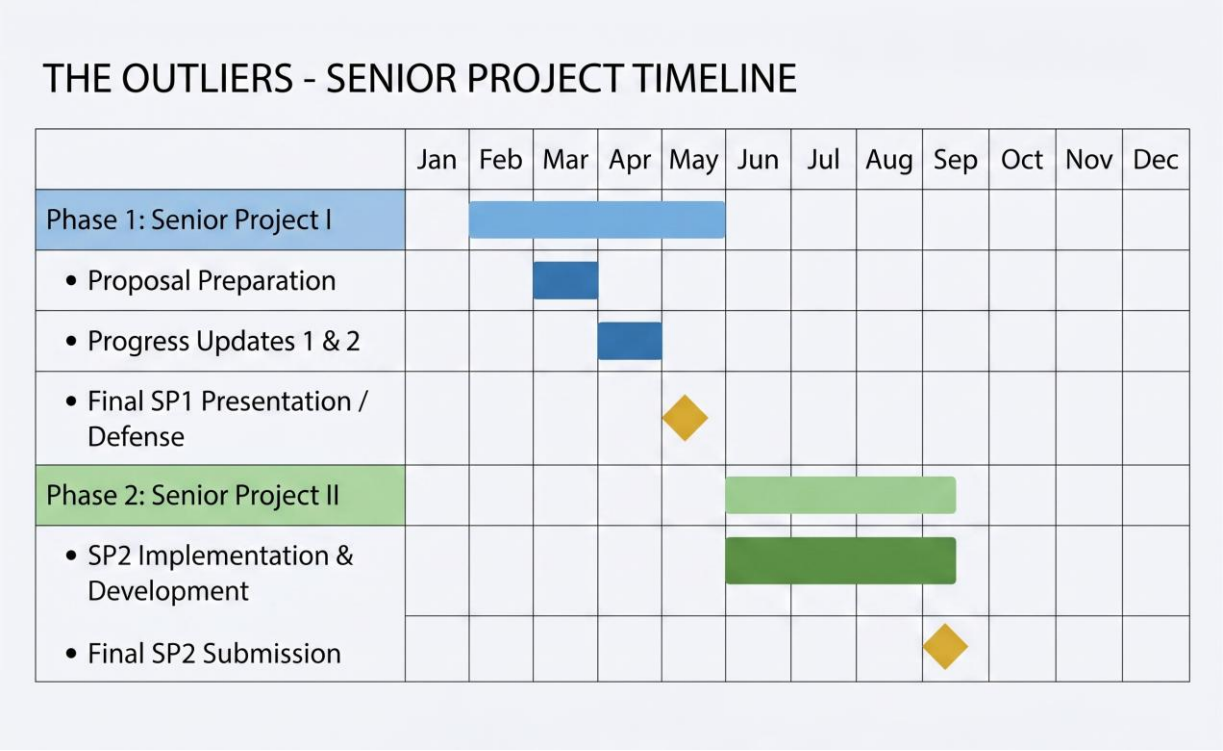
- **Model Development (Phase 2):** Developed, trained, and evaluated four distinct algorithms: XGBoost, Random Forest, SVR, and MLR. Conducted a rigorous comparative analysis using performance metrics, ultimately selecting ***XGBoost*** (achieving an accuracy of R^2 of **86.52%** and an **MAE of 5.38**) as the primary predictive engine due to its superior capability in handling chaotic meteorological data over weaker linear baselines.

  API Integration (Phase 3): Engineered the "on-the-fly" inference pipeline by connecting the OpenWeatherMap API to fetch live weather inputs and pass them through the trained XGBoost model.

- **System Deployment (Phase 4):** Designed and developed the 3-tab Streamlit Web Dashboard featuring a professional user interface.

1.5 Plan

Table 1.1 Working plan



1.6 Expected Results

1.6.1 Software Deliverable (The Product)

- **A Fully Operational "Localized Air Quality Dashboard":** A professional web-based application developed in Streamlit, capable of visualizing real-time PM2.5 levels, dynamic 5-day trend charts, and color-coded health guidelines.

  Live API Integration: A functioning, on-the-fly data pipeline that successfully fetches and processes meteorological ingredients from the OpenWeatherMap API.

1.6.2 Analytical Deliverable (The Intelligence)

- **Validated Predictive Engine:** An advanced Machine Learning architecture evaluating four different algorithms. The system demonstrates a quantifiable accuracy score (peaking at an R^2 of 86.52% via XGBoost), trained on 7 years of highly reliable historical data.

- **Live Model Showdown:** A distinct analytical feature demonstrating real-time, four-way prediction comparisons between the advanced ensemble models (XGBoost, Random Forest) and the traditional baseline models (SVR, MLR) directly within the dashboard.

1.6.3 User Impact (The Benefit)

- **Hyper-Local Accuracy:** Unlike global applications that average data over large geographical grids, this system calculates specific air quality predictions tailored to the unique topography of the Mae Fah Luang University campus.

- **Proactive Health Protection:** Users will receive actionable insights and a 5-day future outlook, allowing them to make informed decisions to reduce respiratory health risks.

1.7 Place and Equipment

1.7.1 Development Location

- **Primary Site**: Mae Fah Luang University, specifically utilizing the Computer Engineering Labs for technical infrastructure and the University Library for documentation and research.

1.7.2 Hardware Specifications

- Device: Laptop Computer (Workstation configuration). Operating System: Windows 11 (64-bit).
  Processor (CPU): Intel Core i5/i7 or AMD Ryzen 5/7 (Minimum 4 Cores).

- Memory (RAM): Minimum 16GB DDR4.

- Storage: 512GB SSD.

1.7.3 Software Specifications

- **Integrated Development Environment (IDE):** Visual Studio Code (VS Code).

- **Programming Language:** Python 3.10+.

- **Frontend Framework:** Streamlit (for building the Web Dashboard).

- **Machine Learning Library: XGBoost** (for Gradient Boosting modeling) and **ScikitLearn** (for data scaling and performance evaluation metrics).

- **Data Manipulation:** Pandas and NumPy.

- **API Service: OpenWeatherMap API** (for real-time and forecasted meteorological data retrieval).

CHAPTER 2 LITERATURE REVIEW

2.1 Related Theory

Temperature Inversion (The Valley Effect): Normally, warm air rises, carrying smoke away. In a valley (like MFU), a layer of warm air can sit on top of cold air near the ground, acting like a "lid" that traps PM2.5. Our model focuses on 8 meteorological variables (Temperature, Wind Speed, Humidity, etc.) specifically to detect this phenomenon.

Ensemble Machine Learning: This project utilizes "Ensemble" methods, which combine multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. This study evaluates various approaches, including:

- **Regression-based Baselines:** To establish a starting accuracy level using linear relationships.

- **Extreme Gradient Boosting (XGBoost):** An advanced gradient-boosted decision tree (GBDT) framework. It optimizes an objective function containing a training loss term and a regularization term :

$$Obj(\Theta) = L(\theta) + \Omega(\Theta)$$

This mathematical structure allows the system to capture complex, non-linear relationships in chaotic weather shifts while preventing "overfitting," ensuring accurate pollution forecasts for the MFU topography.

2.2 Related Research

1. **Global Grid Limitations:** Traditional global models (like those used by AirVisual) typically use a coarse 9km x 9km grid. This resolution is too low to capture the specific airflows and microclimates of small valley basins like the MFU campus, highlighting the need for hyper-local forecasting.

2. **Micro-Climate Similarity:** Studies indicate that Chiang Mai and Chiang Rai share identical Northern Thailand valley basin characteristics. This validates using a continuous 7year dataset from Chiang Mai for model training, which compensates for highly fragmented local MFU data.

3. **Machine Learning Efficacy:** Research shows that traditional linear models struggle with chaotic weather data. Ensemble algorithms like XGBoost and Random Forest significantly outperform them by successfully capturing the complex, non-linear relationships between sudden weather shifts and PM2.5 accumulation.

4. **Meteorological Drivers:** Environmental studies confirm that key weather parameters (Temperature, Humidity, Wind Speed, and Pressure) primarily drive PM2.5 dispersion in valleys. This scientifically justifies using these specific meteorological features to establish our predictive baseline.

2.3 Related Work

- **IQAir (AirVisual)**:

  - Uses a coarse **9km x 9km grid** which is too broad for the MFU valley's specific micro-climate.
  - Relies on the Global Forecast System (GFS) that often averages out localized pollution spikes.
  - Data is excellent for general awareness but lacks the hyper-local precision needed for campus-level safety.

- **Air4Thai (PCD)**:



  - Provides high-precision, government-validated data but the nearest station is **1520km away** in Mueang Chiang Rai.
  - Focuses on **reactive reporting** of current levels rather than providing a proactive 5day predictive forecast.

- Does not account for the "bowl effect" or temperature inversions specific to the MFU campus topography.

- **OpenWeatherMap API**:



- Serves as a robust source for real-time meteorological data but its native AQI models are often generalized.
- Satellite-based estimations can suffer from **"nighttime blindness"** or cloud-cover interference.
- Our project uses their raw weather data as input for a **custom-trained ML engine** to achieve localized accuracy.

2.4 Related Technology

1. Python (Pandas & Scikit-Learn)

Chosen for its robust data manipulation capabilities. Pandas is used to clean the raw Kaggle CSV data, while Scikit-Learn handles feature scaling and model evaluation metrics.



2. Streamlit

A specialized Python library used to develop the proactive web dashboard. It enables the rapid deployment of an interactive user interface featuring real-time AQI updates and 5day predictive trend charts without requiring extensive HTML/CSS.



3.OpenWeatherMap API

A cloud-based API utilized to dynamically fetch live and 5-day forecasted weather data for MFU's exact GPS coordinates. This replaces static databases, enabling real-time, onthe-fly PM2.5 predictions



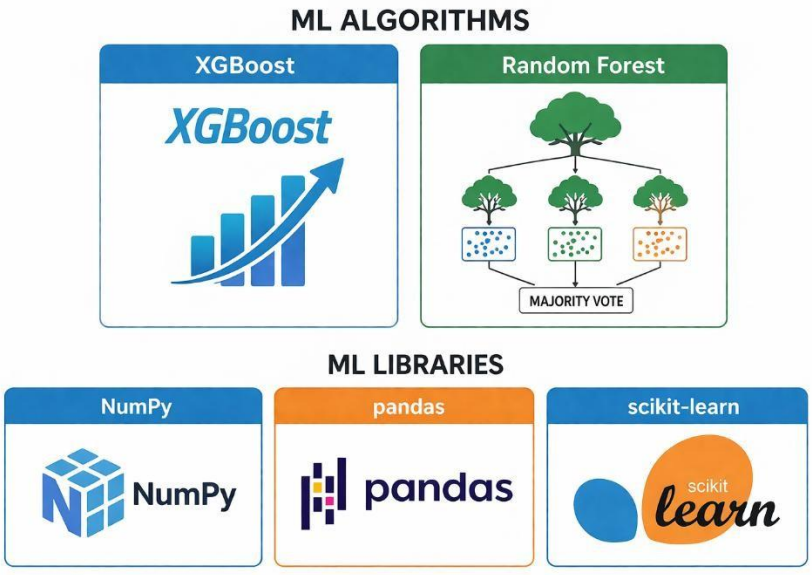
4. Algorithms & Libraries

XGBoost, Scikit-Learn

- **XGBoost:** The champion gradient boosting engine used for high-performance PM2.5 forecasting.
- **Scikit-Learn:** Utilized for critical preprocessing tasks, including feature scaling, data splitting, and calculating performance metrics like R^2 and MAE.

Pandas & NumPy

Core libraries used for data manipulation and numerical analysis. **Pandas** is specifically employed to clean and structure the 7-year historical training dataset from CSV sources into a machine-learning-ready format.



5. MySQL (XAMPP)



The relational database management system used to store and manage historical pollution logs. It serves as the data storage layer for the system's ETL pipeline, ensuring continuous record-keeping for future model retraining.

CHAPTER 3 ANALYSIS AND DESIGN

3.1 Existing System

Currently, the MFU community relies on general-purpose global applications like AirVisual. The typical process flow is inherently reactive: users generally only check the application after physically detecting or smelling smoke. Furthermore, the AQI data provided by these global applications is often sourced from distant provincial stations (sometimes up to 10-20km away) rather than reflecting the exact hyper-local microclimate of the campus. The fundamental flaw in this architecture is its reactive nature; students are already exposed to and breathing hazardous particulate matter before they seek digital confirmation or take preventative measures.

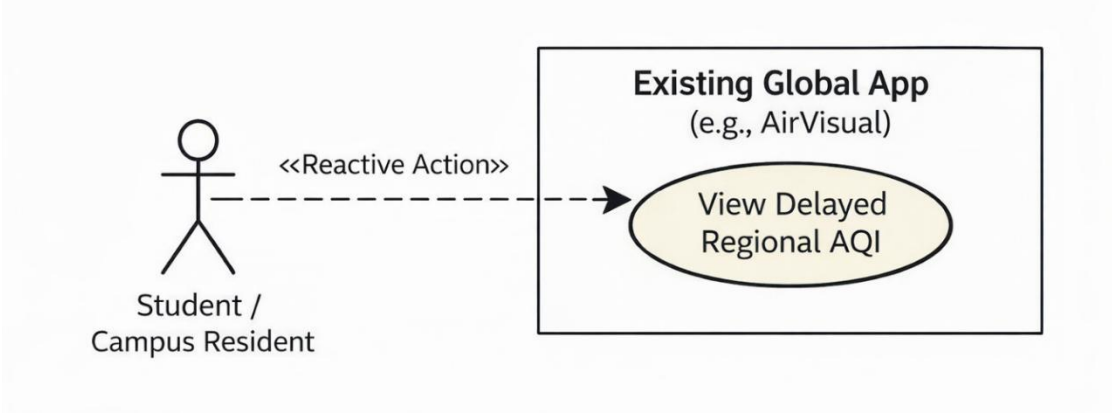


Figure 3.1: Existing System Architecture

3.2 New System

The new system is designed to be "Proactive". Instead of merely displaying current pollution levels, it analyzes atmospheric patterns to warn the student before they even leave their dorm. To achieve this, the architecture is designed as an agile, API-driven 3-Tier Architecture:

- **Presentation Tier:** A Streamlit Web Dashboard featuring a clean user interface, dynamic trend charts, and a dedicated tab for model performance analysis.

- **Logic Tier:** A Python-based application server that handles API requests, preprocesses incoming data via an ETL pipeline, and executes the **pre-trained Machine Learning predictive models**. The system concurrently evaluates multiple algorithms (including XGBoost, Random Forest, Support Vector Regression, and Multiple Linear Regression) to ensure the highest accuracy in capturing the complex, non-linear weather dynamics of the valley.

- **Data Tier:** A hybrid data architecture where the OpenWeatherMap API serves as the dynamic data provider for real-time inference, while a local MySQL database (pm25_project) stores the historical meteorological logs for model training and future continuous learning.

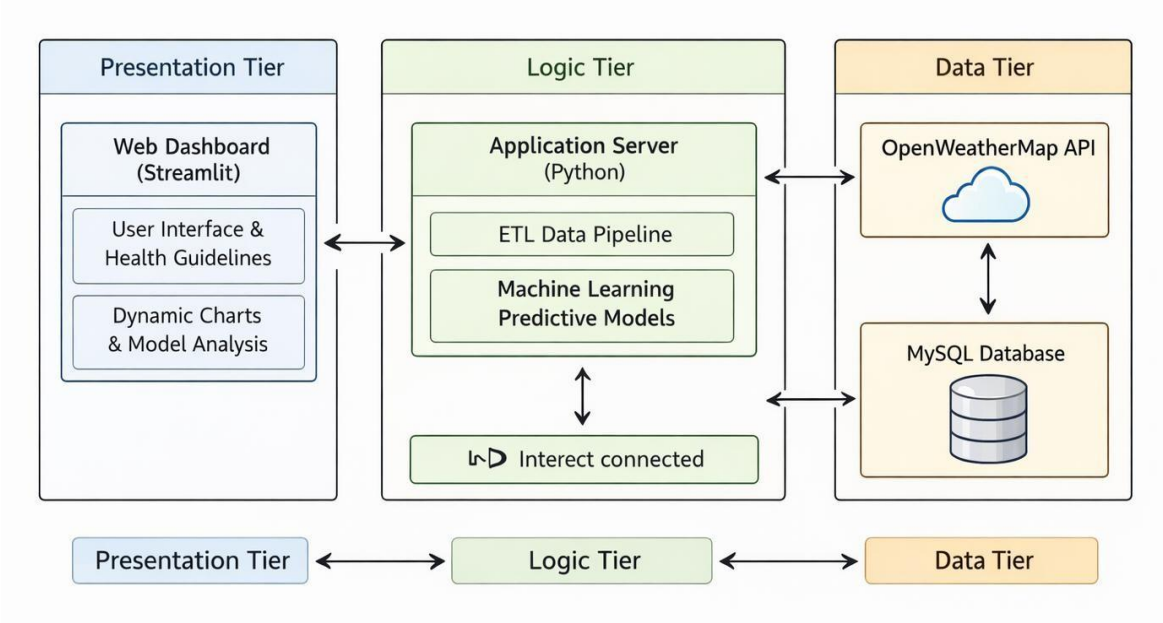


Figure 3.2: New System Architecture

3.3 Use Case Diagram

After gathering user requirements and defining the prototype system's scope, the interactions are streamlined to separate public frontend access from backend system administration:

- **General User (MFU Students & Staff):** Interacts exclusively with the public-facing Streamlit dashboard to **View Real-time AQI**, **Check 5-Day Forecast**, and view comparative model analysis.

- **System Administrator:** Interacts with the backend infrastructure of the application. Their responsibilities include **Managing ML Models** (e.g., adding or updating .pkl model files in the system core), **Executing the Data Pipeline** (e.g., running the ETL scripts to migrate local data into the MySQL database), and **Monitoring API Status** to ensure the continuous flow of OpenWeatherMap data.

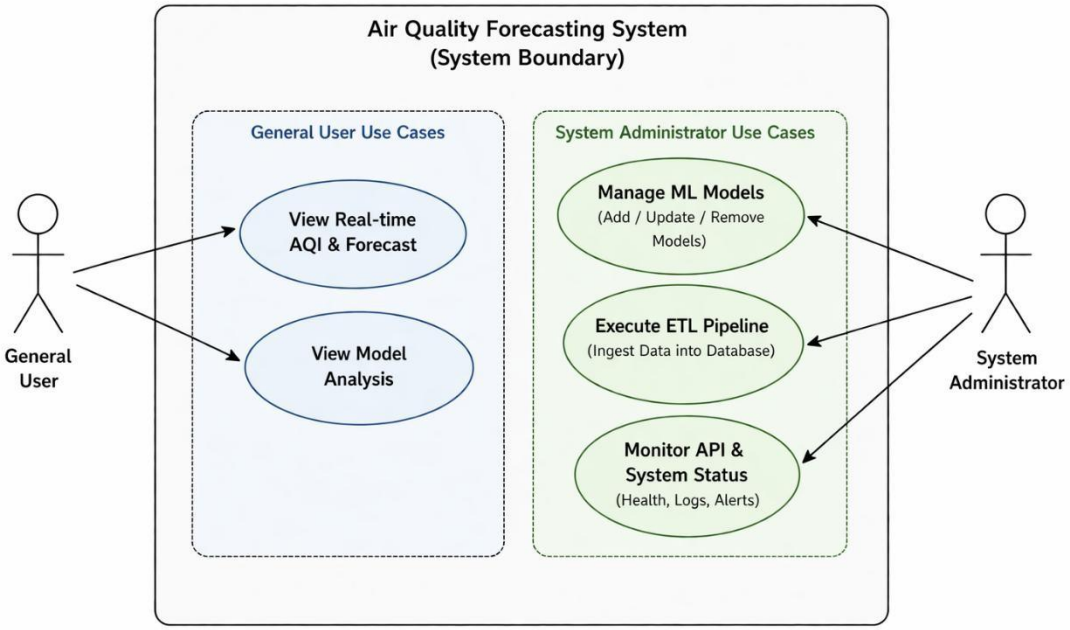


Figure 3.3: Use Case Diagram of System

This figure illustrates these streamlined interactions. This design ensures immediate public accessibility on the frontend while restricting critical data and model management tasks to the backend administrator.

3.4 Activity Diagram

The activity flow of the new "on-the-fly" inference system is significantly streamlined and handles multiple data pipelines concurrently based on the user's dashboard interaction:

1. The user accesses the Streamlit web dashboard via a browser.

2. The system simultaneously executes two data retrieval processes:
    - **For Live & Model Analysis:** It triggers API calls to OpenWeatherMap to fetch live meteorological forecasts and current air pollution data.
    - **For Historical Trends:** It loads preprocessed local environmental data directly from the system's local database.

3. For the live inference pipeline, the system extracts, cleans, and formats the retrieved API data into the 8 required features (7 weather parameters + 1 historical PM2.5 lag).

4. The pre-trained Machine Learning models (including the primary XGBoost engine and baseline algorithms) process these live features to calculate future PM2.5 concentrations.

5. The dashboard dynamically updates the user interface, displaying real-time predictions, comparative model analysis, and interactive historical trend charts across its three dedicated tabs.

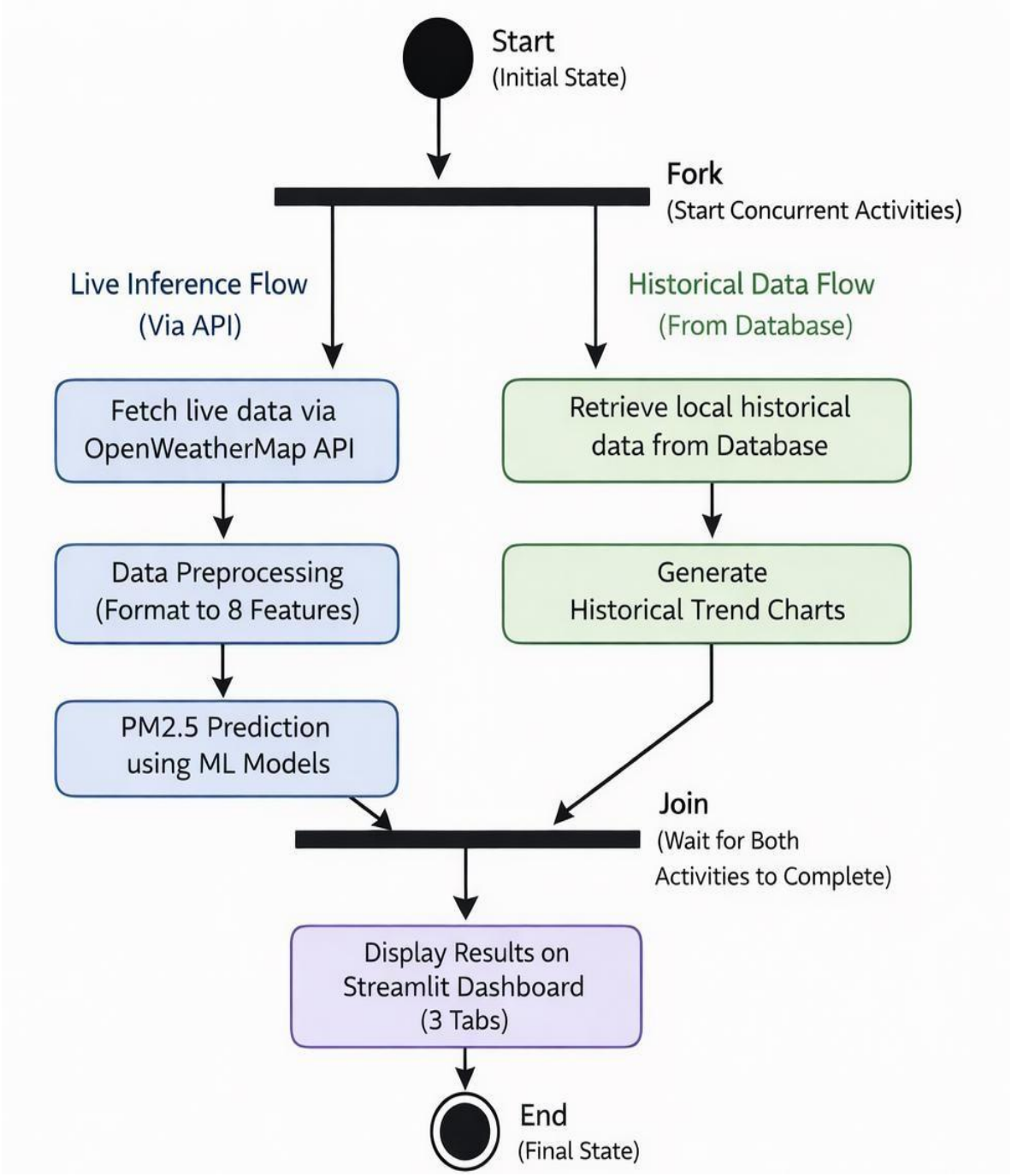


Figure 3.4: Activity Diagram of Dashboard

3.5 Entity-Relationship Diagram (ER)

This figure illustrates the database schema designed for the system's ETL pipeline. The architecture uses MySQL to store preprocessed historical data before it is fed into the machine learning models. The database (pm25_project) currently consists of two entities:

- **pollution_logs**: This table stores the continuous time-series meteorological data along with the PM2.5 target variable. It acts as the primary data source for the historical analysis and machine learning training processes.
    - *Attributes:* **log_date (Primary Key), pressure_avg, temp_avg, humidity_avg, precipitation, sunshine, wind_direct, wind_speed, pm25_lag1, pm25.**

- **users**: This table is initialized in the schema to establish a structural foundation for future system administration and basic role management. For the current prototype phase, it is kept as a minimal entity without requiring complex web authentication or profile management. ○ *Attributes:* user_id (Primary Key).

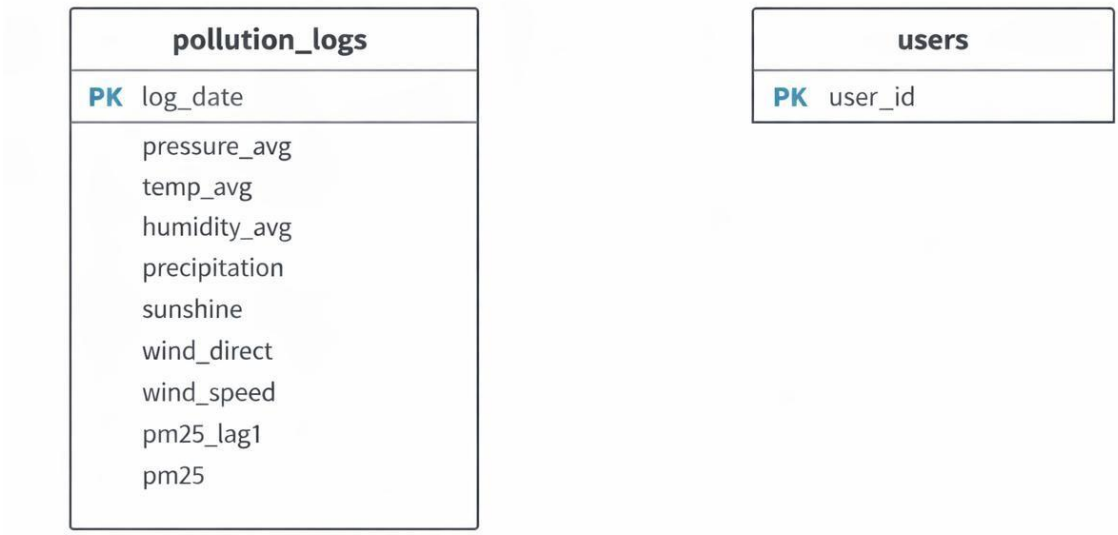


Figure 3.5: ER Diagram of PM2.5 Prediction System

3.5.1 Data Dictionary

To ensure strict data typing and clear engineering standards, the following data dictionaries define the exact structure, data types, and specific units of the entities within the pm25_project database.

Table 3.1: Data Dictionary for pollution_logs Table

Attribute (Field Name)	Data Type	Key	Description	Unit / Example
log_date	DATETIME	PK	Timestamp of the recorded environmental data	YYYY-MM-DD HH:MM:SS
pressure_avg	FLOAT	-	Atmospheric pressure at sea level	hPa (e.g., 1012.5)
temp_avg	FLOAT	-	Average ambient temperature	°C (e.g., 32.5)
humidity_avg	FLOAT	-	Relative atmospheric humidity	% (e.g., 45.0)
precipitation	FLOAT	-	Volume of rainfall	mm (e.g., 0.0)
sunshine	FLOAT	-	Duration of sunshine exposure	Hours (e.g., 5.0)
wind_direct	FLOAT	-	Wind direction angle	Degrees (e.g., 180)
wind_speed	FLOAT	-	Speed of the wind	m/s (e.g., 2.5)
pm25_lag1	FLOAT	-	Previous day's PM2.5 concentration (Historical lag)	$\mu g/m^3$
pm25	FLOAT	-	Target Variable; Current actual PM2.5 level	$\mu g/m^3$

Table 3.2: Data Dictionary for users Table

Attribute (Field Name)	Data Type	Key	Description	Unit / Example
user_id	INT	PK	Unique identifier for the system administrator	Auto Increment (e.g., 1, 2)

3.6 User Interface Design and Value Proposition

The presentation tier is actively developed using Streamlit. Instead of using generic mockup images, the following screenshots demonstrate the actual working prototype of our system. The dashboard is divided into three functional tabs, each designed to deliver a specific value to the MFU community:

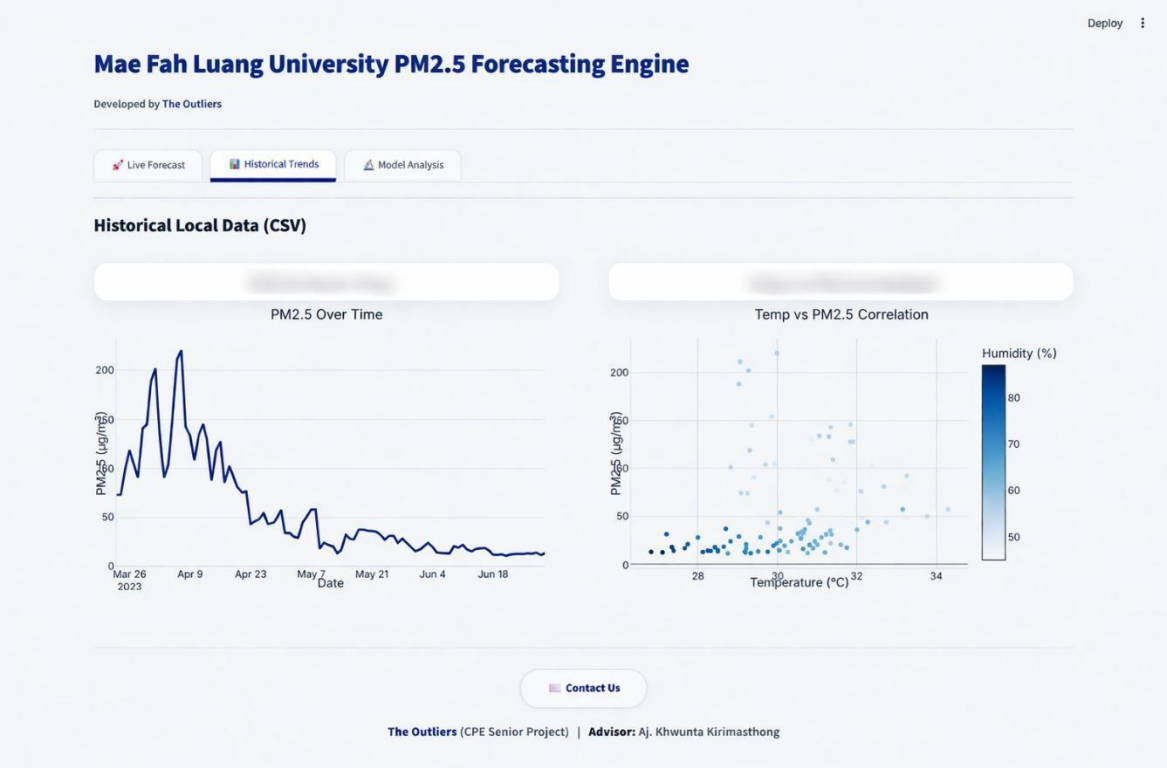
1. Live Forecast Interface (Value: Proactive Health Protection)



- **Description:** This primary view displays the real-time PM2.5 concentration alongside its corresponding color-coded health status (Good, Moderate, Unhealthy, Hazardous). It also features a dynamic 5-day predictive line chart.

- **Delivered Value:** It shifts the community's behavior from "reactive" to "proactive". Students do not have to wait until they smell smoke; they can check the 5-day trend and plan their outdoor activities or prepare N95 masks in advance, effectively reducing respiratory health risks.

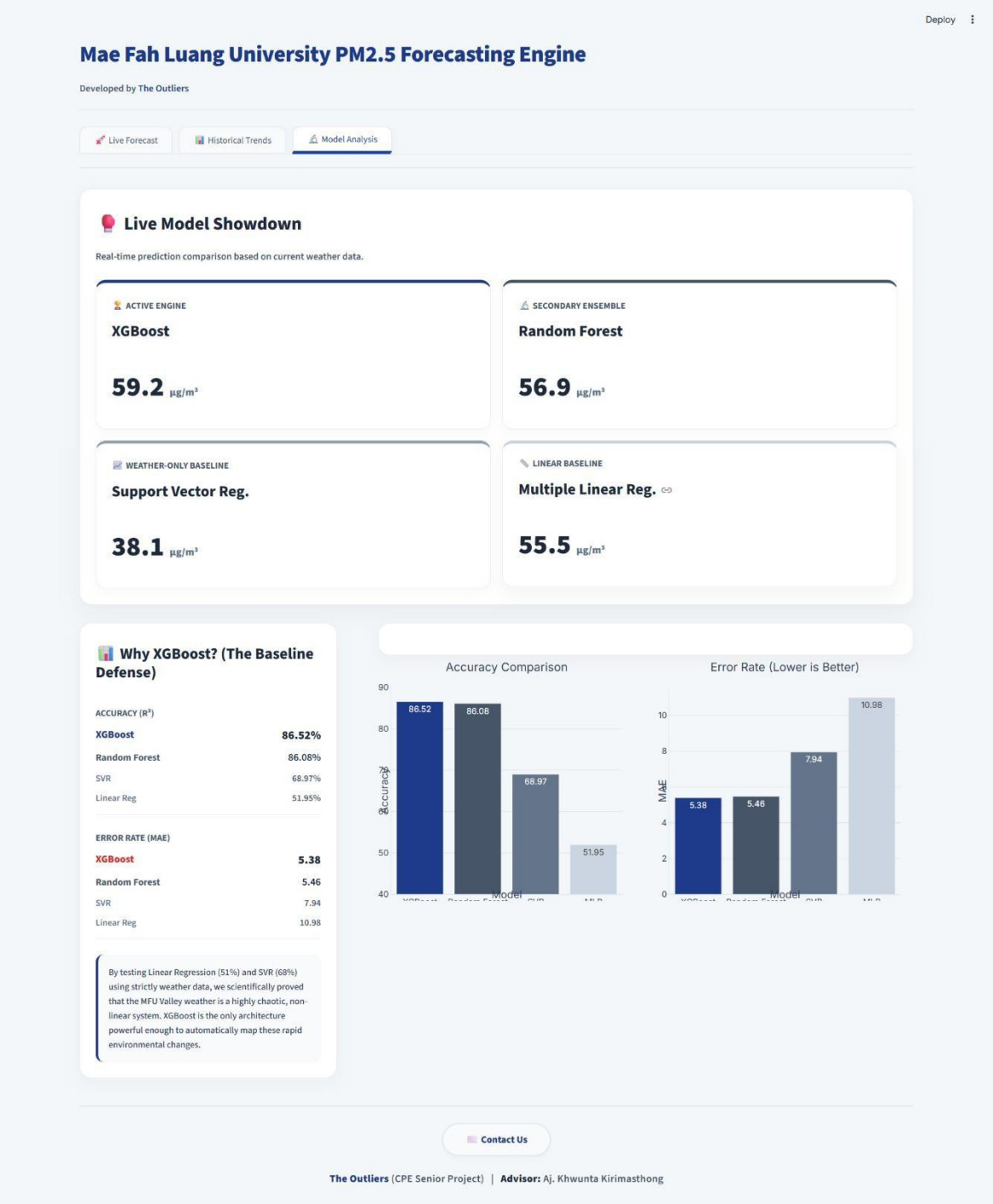
2. Historical Trends Interface (Value: Environmental Education)



- **Description:** This analytical view utilizes local CSV data to render interactive time-series graphs and scatter plots. It visualizes the correlation between temperature, humidity, and pollution levels.

- **Delivered Value:** It educates the campus residents about the "Valley Effect". By visually proving how specific weather shifts (like sudden temperature drops) cause pollution spikes, it builds public awareness regarding Northern Thailand's unique climate physics.

3. Model Analysis Interface (Value: Engineering Transparency & Trust)



- **Description:** This tab features a "Live Model Showdown" displaying realtime predictions from all four evaluated models concurrently (XGBoost, Random Forest, SVR, and MLR). It includes comparative accuracy (R^2) and error rate (MAE) charts.

- **Delivered Value:** It builds user trust through transparency. By openly displaying how the XGBoost engine outperforms traditional linear models, users (and system administrators) can mathematically verify that the daily health warnings are generated by a highly accurate, scientifically validated AI, rather than basic estimations.

CHAPTER 4 CONCLUSION

4.1 Conclusion

This project delivers a specialized, data-driven solution to the hazardous haze crisis in Northern Thailand by shifting the strategic focus from "Global Monitoring" to "Hyperlocal Forecasting." During the development of Senior Project I, the research successfully established a dynamic, on-the-fly inference architecture that bridges the gap between raw environmental data and actionable community insights.
Key technical milestones achieved include:

- **API-Driven Infrastructure:** The seamless integration of the OpenWeatherMap API for fetching real-time and 5-day forecasted meteorological data specific to the MFU GPS coordinates.

- **Advanced ML Framework:** The development and evaluation of a 4-way machine learning framework. Through rigorous testing, **XGBoost** was validated as the champion engine, achieving a peak accuracy (R^2) of **86.52%** and a low Mean Absolute Error (MAE) of **5.38**, significantly outperforming traditional linear baselines.

- **Interactive Deployment:** The final deployment of a professional-grade web dashboard via the Streamlit framework. The platform empowers the MFU community with real-time AQI updates, historical correlation analysis, and a "Live Model Showdown," providing a 5-day window for proactive health planning.

Ultimately, this project proves that localized topography-aware modeling is essential for accurate air quality prediction in valley basins, providing the MFU campus with a specialized tool to mitigate respiratory health risks before hazardous exposure occurs.

4.2 Discussion

To scientifically validate the selection of the core predictive engine, the system evaluated four distinct machine learning algorithms using the exact same pre-processed dataset (comprising 8 meteorological and historical features). The performance was measured using R-Squared (R^2) for overall accuracy and Mean Absolute Error (MAE) for the error rate.

Table 4.1: Performance Comparison of Evaluated Machine Learning Models

Comparison	Model	Accuracy (R2)	Error Rate (MAE)	Performance Status
XGBoost	Gradient Boosting (Ensemble)	**86.52%**	**5.38**	Champion Model (Selected)
Random Forest	Bagging (Ensemble)	**86.08%**	**5.46**	Strong Alternative
Support Vector Regression	Non-linear Baseline	**68.97%**	**7.94**	Underfitting (Struggles with outliers)
Multiple Linear Regression	Linear Baseline	**51.95%**	**10.98**	Baseline Failed (Cannot capture complexity)

Analytical Justification:

The comparative results *[Table 4.1]* clearly demonstrate the fundamental nature of atmospheric pollution modeling. By testing traditional Multiple Linear Regression (R^2: 51.95%) and Support Vector Regression (R^2: 68.97%) exclusively on weather data, we mathematically proved that the MFU Valley weather is a highly chaotic, non-linear system. Simple linear equations completely fail to capture sudden temperature inversions or shifting wind patterns.

Conversely, the tree-based ensemble models, particularly **Extreme Gradient Boosting (XGBoost)**, successfully mapped these rapid, non-linear environmental changes, peaking at an accuracy of **86.52%** with the lowest error margin (MAE: 5.38). This rigorous 4-way evaluation scientifically justifies why XGBoost is deployed as the active engine for the real-time Streamlit dashboard.

Future Work Plan (Senior Project II)

The preliminary analysis and model evaluation confirm that meteorological factors are strong correlates for PM2.5 fluctuations within the MFU valley topography. While the primary **XGBoost** model demonstrated superior accuracy (R^2 of **86.52%**), we observed that environmental "outliers," such as localized agricultural burning or sudden forest fires, remain a significant challenge for weather-only predictive models.

To address these challenges, future work in Senior Project II will focus on upgrading the architecture to a **"Geographically Aware Big Data Engine"**:

- **NASA FIRMS Integration:** We will **integrate real-time satellite hotspot data** from NASA to account for the actual source of the pollution.

- **Time-Series Memory:** We will introduce "Triple Lags" for both PM2.5 and Fire Pressure to mathematically capture the "Bowl Effect" where smoke lingers in the valley over several days, pushing the accuracy beyond the current baseline.

REFERENCES

[1] Pollution Control Department, "Historical Air Quality Data 2016-2023," Bangkok, Thailand, 2024. [Online]. Available: http://air4thai.pcd.go.th. [Accessed: 06-Feb2026].

[2] IQAir, "AirVisual: World's Air Quality Index (AQI) & Pollution Data," 2024. [Online]. Available: https://www.iqair.com. [Accessed: 06-Feb-2026].

[3] F. Pedregosa *et al.*, "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://scikit-learn.org. [Accessed: 06-Feb-2026].

[4] The Pandas Development Team, "pandas-dev/pandas: Pandas," Zenodo, Feb. 2020. [Online]. Available: https://pandas.pydata.org. [Accessed: 06-Feb-2026].

[5] Streamlit Inc., "Streamlit Documentation," 2024. [Online]. Available: https://docs.streamlit.io. [Accessed: 06-Feb-2026].

[6] C. D. Whiteman, *Mountain Meteorology: Fundamentals and Applications*. New York, NY, USA: Oxford University Press, 2000.

[7] World Health Organization, "Ambient (outdoor) air pollution," 2023. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-airpollution. [Accessed: 06-Feb-2026].